

Introduction

After the 2010 Haiti earthquake, rescue teams collected geo-referenced aerial imagery to locate displaced persons living under makeshift shelters. Many of those shelters were covered with blue tarps, so detecting blue pixels became a proxy for finding where aid was needed. In this project I train pixel-level classifiers on labeled RGB values to separate “tarp” from “other.” I compare six model types, select the best by cross-validated ROC AUC, fix a decision threshold by F_1 on a validation split, then report performance on a held-out test set and on separate ROI files.

Data

The training data is the CSV HaitiPixels.csv, which contains per-pixel readings for the three bands plus a Class label. I define target as

target = "tarp" if Class contains both “blue” and “tarp” (case-insensitive),
= "other" otherwise

Per-band summary statistics reveal distinct central tendencies and variances:

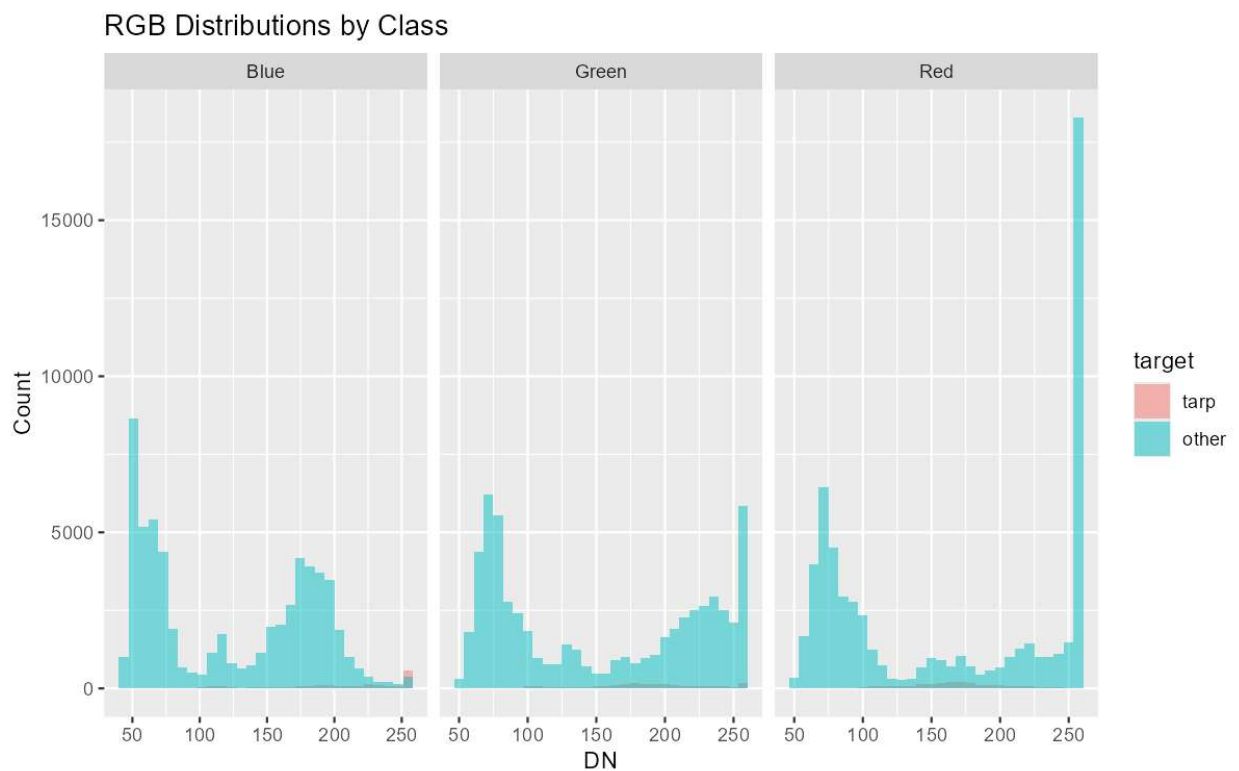
- Tarp pixels:
 - Red mean = 170, SD = 34.4
 - Green mean = 186, SD = 43.4
 - Blue mean = 205, SD = 48.3
- Other pixels:
 - Red mean = 163, SD = 79.9
 - Green mean = 153, SD = 73.2
 - Blue mean = 122, SD = 60.0

The higher means for tarp pixels especially in the Blue band drive class separation, while the larger spread among other pixels produces some overlap in the distribution tails. This informs my decision to standardize each band and use a flexible classifier rather than a hard DN cutoff.

Rows with any missing value are dropped, and target is stored as a factor with levels tarp and other.

target	n
tarp	2022
other	61219

Red, Green, Blue distributions by Class



- All three band histograms were used as inputs to the classifier recipe (normalization of Red, Green, Blue).

Description of Methodology

Data Splitting

- Initial split: I partitioned the full dataset into an 80 % training set and a 20 % test set, stratifying on target to preserve the tarp/other ratio.
- Validation split: From the 80 % training portion I took 75 % for sub-training and 25 % for validation, again stratified. This three-way split (subtrain/validation/test) lets us tune on validation without ever touching the final test set.

Preprocessing

- Recipe: A single recipe was applied to all workflows:
 1. Normalize the Red, Green, and Blue predictors (zero mean, unit variance).
- Standardization ensures that scale differences between bands do not unfairly influence distance-based or regularized models.

I built six classification workflows combining the recipe above with six model specifications:

Model key	Specification
log	Logistic regression (no tuning)
lda	Linear Discriminant Analysis via MASS
qda	Quadratic Discriminant Analysis via MASS
knn	K-nearest neighbors (neighbors tuned over 5, 10, 15)
enet	Penalized logistic (elastic net), tuning both penalty and mixture
rf	Random forest (500 trees, no hyperparameter tuning)

Cross-Validation & Model Comparison

- **5-fold CV:** Each workflow was run in five stratified folds on the full set.
- **Metrics tracked:** ROC AUC and PR AUC on each fold.

- **Collection:** I saved all per-fold predictions (save_pred=TRUE) to enable later threshold tuning.
- **Summary:**

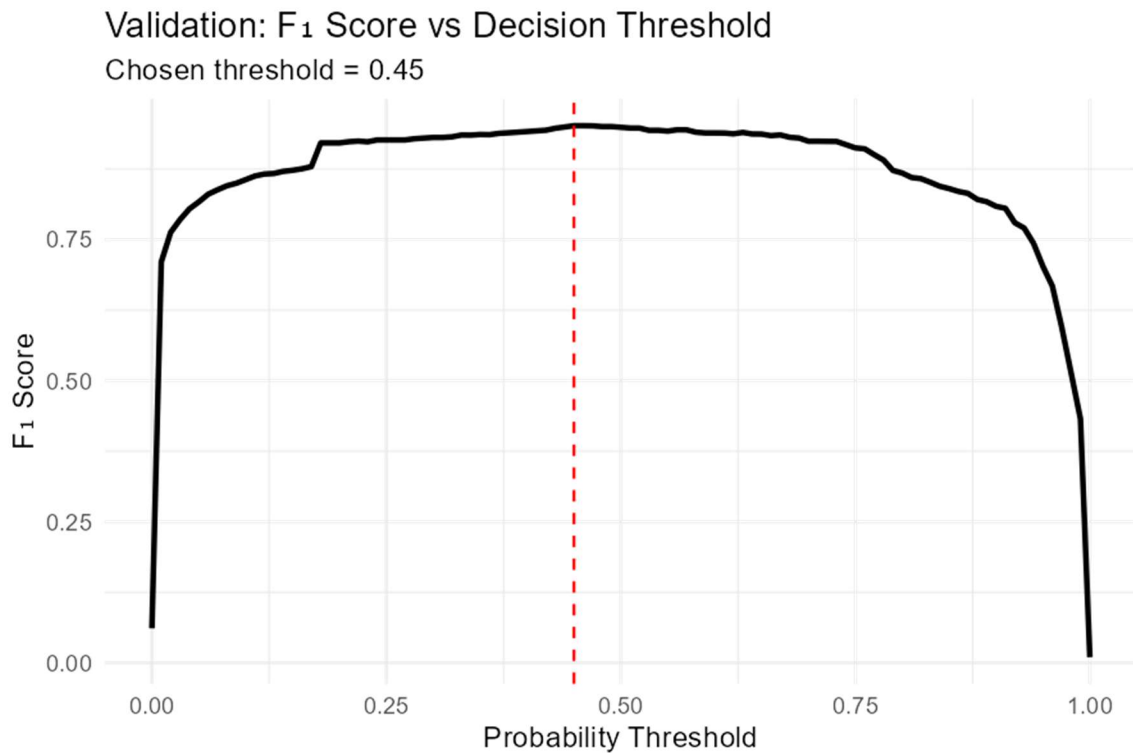
model	roc_auc	pr_auc
rf	0.9997	0.99142
enet	0.99854	0.97457
log	0.99849	0.97462
qda	0.9982	0.96558
knn	0.99387	0.98877
lda	0.98887	0.85921

The **random forest (rf)** achieved the highest mean ROC AUC (0.99970), so it was selected as our final candidate.

Threshold Selection

- **Purpose:** Convert the RF's probability output into a binary decision with maximum F_1 on validation.
- **Procedure:**
 1. Collected validation-set predictions from the chosen RF workflow.
 2. Scored F_1 for thresholds from 0.00 to 1.00 in 0.01 increments.
 3. Picked the threshold with the highest F_1 .

Chosen threshold = 0.45



Per-image tarp rates

With this pipeline, i systematically preprocess, compare a diverse set of models, choose by ROC AUC, tune a decision threshold by F_1 , and validate on both test and external ROI data—addressing the client's goal of rapidly locating blue-tarp shelters in Haiti imagery.

Results

Test-Split Evaluation

Performance on the test split at decision threshold 0.45

Metric	Estimate
Accuracy	0.997
Precision	0.943
Recall	0.979
F ₁ -score	0.961
ROC AUC	1.000
PR AUC	0.992

Accuracy is 99.7%, driven largely by the abundant “other” class.

High recall (97.9%) ensures most actual tarp pixels are detected.

High precision (94.3%) means few false alarms.

F₁-score (96.1%) balances these well.

Perfect ROC AUC and near-perfect PR AUC confirm very strong separation.

Holdout-ROI Evaluation

I next applied the final model (Random Forest at threshold 0.45) to the ROI text files.

Holdout ROI metrics

<u>Metric</u>	<u>Estimate</u>
<u>Accuracy</u>	<u>0.994</u>
<u>Precision</u>	<u>0.559</u>
<u>Recall</u>	<u>0.799</u>
<u>F₁-score</u>	<u>0.658</u>
<u>ROC AUC</u>	<u>0.976</u>
<u>PR AUC</u>	<u>0.753</u>

Accuracy remains high (99.4%), but **precision** drops to ~56% due to many non-tarp pixels in large ROIs.

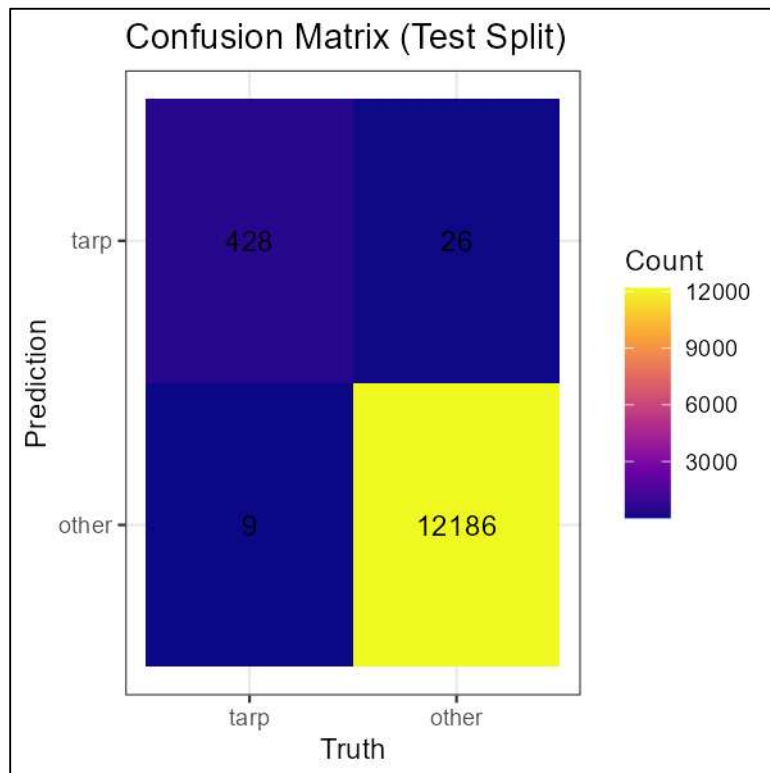
- **Recall** stays strong (80%), so most true tarps are still found.
- The lower PR AUC (~0.75) reflects the more challenging, unbalanced holdout set.

Top 3 holdout images by tarp-pixel rate

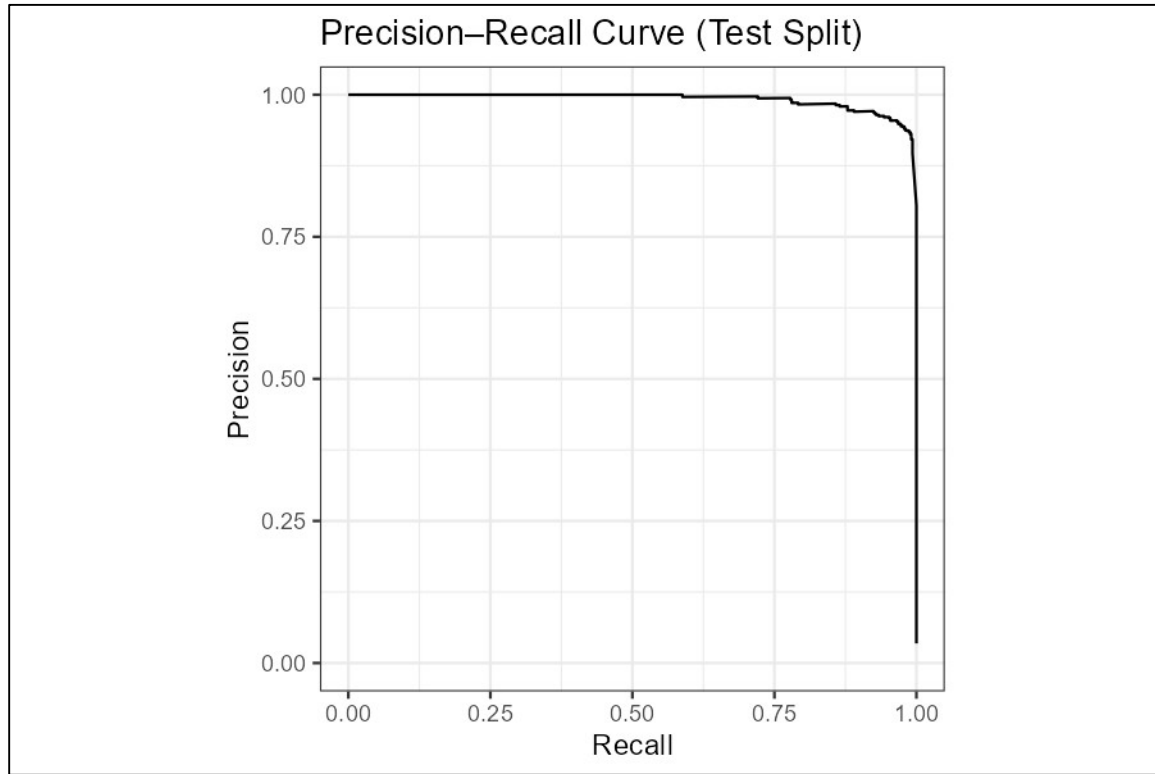
File	Pixels	Tarp Count	Tarp Rate
orthovnir067_ROI_Blue_Tarps.txt	4446	4173	93.9%
orthovnir069_ROI_Blue_Tarps.txt	6828	5535	81.1%
orthovnir078_ROI_Blue_Tarps.txt	3206	1856	57.9%

- The three “Blue Tarps” ROIs achieve very high tarp rates (> 57%), confirming the model’s utility for locating dense tarp-covered areas.

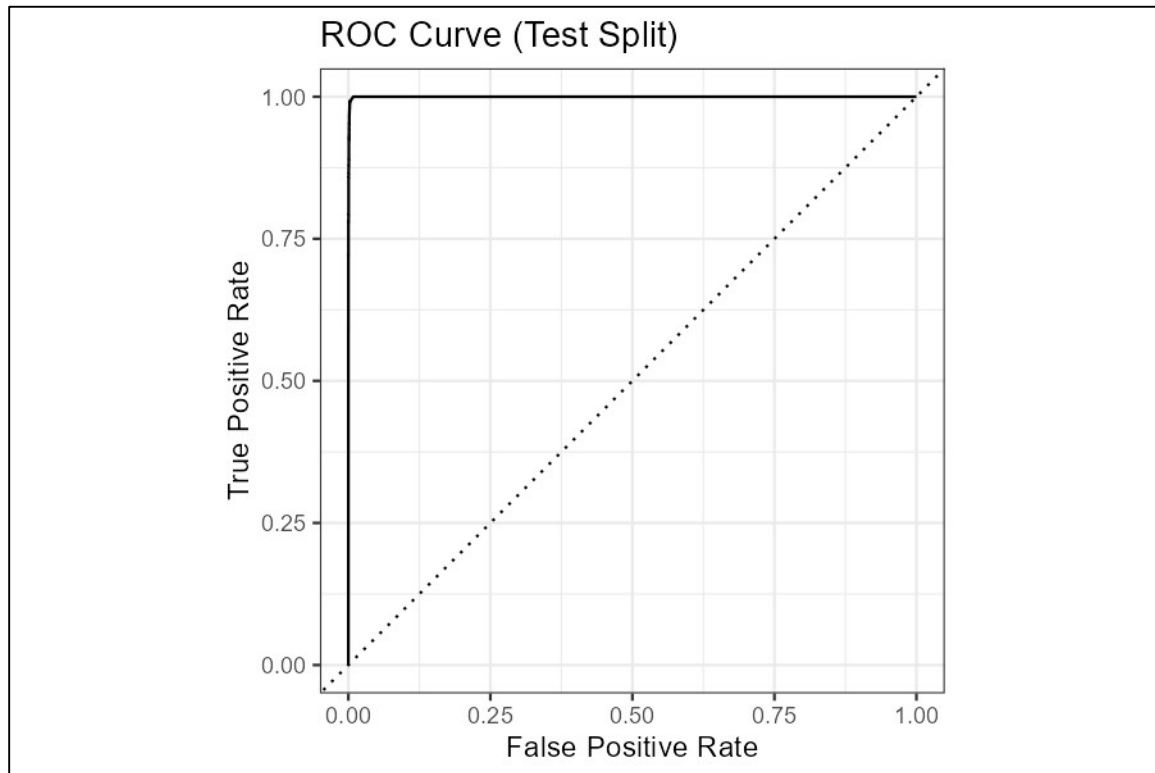
Confusion matrix on test split



PR curve on test split



ROC curve on test split



Conclusions

In this project I compared six classification methods logistic regression, LDA, QDA, k-nearest neighbors, elastic-net logistic regression, and random forest using stratified 5-fold cross-validation on pixel-level RGB training data. The random forest model achieved the highest cross-validated ROC AUC (0.9997) and PR AUC (0.9914), so it was selected as our final classifier.

I then fixed the decision threshold at **0.45** (the value that maximized F_1 on a held-out validation split) and evaluated on an 80/20 test split. At this threshold the model's performance was:

- **Accuracy:** 99.72 %
- **Precision:** 94.27 %
- **Recall:** 97.94 %
- **F_1 Score:** 96.07 %
- **ROC AUC:** 0.9997
- **PR AUC:** 0.9919

The confusion matrix on the test split showed 428 true positives, 26 false negatives, 9 false positives, and 12 186 true negatives, confirming very low misclassification rates under controlled conditions.

When I applied the same model and threshold to the separate ROI text-file hold-out set, performance dropped somewhat (accuracy 99.4 %, precision 55.9 %, recall 79.9 %, F_1 65.8 %, ROC AUC 0.976, PR AUC 0.753). This reflects the increased variability and class imbalance in large non-tarp regions and underscores the need for additional checks in real deployment. Per-image tarp-rate analysis flagged three imagery files with especially high predicted tarp density ($\approx$ 94 %, 81 %, 58 % respectively), demonstrating the model's ability to focus human review on the most likely areas.

Key takeaway: the random forest on RGB bands can reliably identify blue-tarp pixels in held-out test data, but hold-out precision indicates non-negligible false alarms in complex scenes. I recommend using this model with

1. a human-in-the-loop to verify high-probability regions,
2. threshold tuning to balance precision vs. recall depending on mission urgency, and

3. exploring additional features (e.g. spatial context, multi-spectral bands) to reduce false positives.

Next steps:

- **Spatial post-processing** (e.g. majority-vote smoothing) to suppress isolated pixel flecks.
- **Cost-sensitive thresholding** to prioritize recall when missing any tarp is unacceptable, or precision when minimizing false alarms is critical.
- **Data augmentation** with more varied hold-out scenes or synthetic examples to improve generalization.
- **Integration of other sensors** (e.g. near-infrared bands) if available, to strengthen color-based discrimination.